Incremental Data-Driven Robust Crew Pairing Optimization

Under Evolving Uncertainty

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Motivation $\rightarrow$ Uncertainty model $\rightarrow$ Robust optimization $\rightarrow$ Re-optimization

Why Crew Pairing Is Difficult

Planning requirement

- Every flight must be covered.
- Crew duties must satisfy rules.
- Pairing cost should be controlled.

Operational uncertainty

- Delays break flight connections.
- Recovery creates extra cost.
- Rare disruptions are important.

Project goal

Build a crew pairing plan that is low-cost, robust to delays, and easy to update when new data arrive.

Data-Driven Uncertainty Set

Why not only fit a distribution?

- Limited disruption data.
- Correlation and extreme shifts are hard to capture.
- Normal days can dilute serious delay events.

Proposed idea

- Use empirical quantiles from delay data.
- Model only upward delay deviations.
- Control conservatism through a budget.

$$d_f = \mathrm{Quantile}_\alpha \left(x_f^1, x_f^2, \ldots, x_f^T \right), \qquad 0 \le \Delta_f \le d_f$$

Balancing Robustness And Cost

Small Γ
lower cost $\rightarrow$ Budget parameter
$\sum_f \Gamma_f \leq \Gamma$ $\rightarrow$ Large Γ
higher protection

$$\hat{\mathcal{U}}(\Gamma) = \left\{ \Delta : 0 \leq \Delta_f \leq d_f,\ \sum_f \Gamma_f \leq \Gamma,\ \Delta_f = \Gamma_f d_f \right\}$$

- Protects against serious delay scenarios.
- Avoids assuming all flights are delayed at once.
- Lets the airline tune the level of conservatism.

Two-Stage Robust Optimization Model

First stage: planning

Choose crew pairings before the actual disruption is known.

$$\min_{x} \sum_{p \in P} c_p x_p + \theta$$

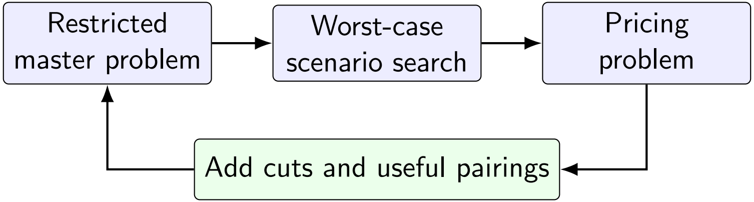
Second stage: recovery

Under delay scenarios, choose recovery actions.

- Wait and absorb delay.
- Swap crew or connection.
- Cancel when necessary.

$$\min_{x} \left\{ \text{planned cost} + \max_{\Gamma \in \hat{\mathcal{U}}} \text{recovery cost}(x, \Gamma) \right\}$$

Solution Algorithm



- Column generation handles the large pairing space.
- Nested C&CG handles mixed-integer recourse and worst-case scenarios.
- The algorithm stops when no useful pairing or violated scenario remains.

Incremental Re-Optimization

When uncertainty shrinks

- New bounds are tighter.
- Old robust solution remains feasible.
- The plan is conservative but safe.

When uncertainty expands

- New bounds are looser.
- Add new robustness cuts.
- Re-optimize without rebuilding from zero.

Practical value

The model can adapt when operational data evolve, while reusing the previous optimization structure.

Conclusion

- Crew pairing needs schedules that are feasible, low-cost, and robust.
- Quantile-based uncertainty sets use historical delay data directly.
- Two-stage robust optimization models planning and recovery together.
- Nested C&CG and column generation solve the model iteratively.
- Incremental re-optimization updates the model as uncertainty evolves.

Final takeaway

A robust, data-driven, and adaptive framework for airline crew pairing under evolving delay uncertainty.